

Technical Assessment Report: Secure Cloud Deployment of PrestaShop E-Commerce Platform

Engineer: Msugh Felix Demkiir

Project: Preliminary Assignment for Cyber Security Internship

Application URL: <http://13.62.103.215/index.php>

Secure Admin Portal: <http://13.62.103.215/admin244izxrcfqftd2dd9vg/>

Project Overview

This report details the deployment of a professional PrestaShop e-commerce ecosystem hosted on Amazon Web Services (AWS). The primary goal of this architecture was to move away from a "monolithic" setup (where everything is on one server) and instead build a decoupled, multi-tier infrastructure.

By hosting the web server on EC2 and the database on a managed RDS instance, I have created a system that is:

- **Highly Secure:** The database is isolated in a private tier and is not reachable via the public internet.
- **Performant:** The application and data tiers have dedicated resources, preventing "resource contention."
- **Scalable:** This setup allows for independent scaling of the web and database layers as store traffic grows.

Objective

To design, deploy, and secure a decoupled, cloud-based e-commerce infrastructure utilizing AWS EC2 and RDS. The deployment emphasizes network isolation, secure

system administration, and the implementation of a LAMP stack tailored for PrestaShop.

Phase 1: Database Backend Provisioning (AWS RDS)

To ensure high availability and separate the data tier from the application tier, a dedicated managed database was provisioned.

- **Service:** Amazon Relational Database Service (RDS)
- **Engine:** MySQL Community Edition
- **Instance:** prestashop-db (db.t4g.micro)
- **Credentials:** User: admin | Password: [REDACTED FOR SECURITY]
- **Action:** Successfully initialized the database and captured the internal AWS Endpoint for secure internal routing.

Engine options

Engine type [Info](#)

☒ Aurora (MySQL Compatible)



☐ Aurora (PostgreSQL Compatible)



☐ MySQL



☐ PostgreSQL



☐ MariaDB



☐ Oracle



☐ Microsoft SQL Server



☐ IBM Db2

Choose a database creation method

☒ Full configuration

You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create

Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Templates

Choose a sample template to meet your use case.

☒ Production

Use defaults for high availability and fast, consistent performance.

☐ Dev/Test

This instance is intended for development use outside of a production environment.

☒ Free tier

Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS.

Connection details to your database prestashop-db

This is the only time you can view this password. Copy and save the password for your reference. If you lose the password, you must modify your database to change it. You can use a SQL client application or utility to connect to your database.

[Learn about connecting to your database](#)

Master username

admin

Master password

Endpoint

prestashop-db.cnaqkk2lszib.eu-north-1.rds.amazonaws.com

Close

Phase 2: Web Server Provisioning (AWS EC2)

A virtual machine was spun up to serve as the public-facing application host.

- **Service:** Amazon Elastic Compute Cloud (EC2)
- **OS Image:** Ubuntu Server 24.04 LTS
- **Instance Type:** t3.micro (**Prestashop-Web**)
- **Security Group Configuration:** Created **Prestashop-Web-SG**.
 - Opened **Port 80 (HTTP)** and **Port 443 (HTTPS)** to **0.0.0.0/0** for public web traffic.
 - Restricted **Port 22 (SSH)** strictly to the administrator's local IP address.

Quick Start

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), 55D Volume Type
ami-080254318c2d8932f (64-bit x86) / ami-0af74201b53556fe77 (64-bit Arm)
Virtualization: hvm ENA enabled: true Root device type: ebs

Free tier eligible

Description

Ubuntu Server 24.04 LTS (HVM),EB5 General Purpose (55D) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).
Canonical, Ubuntu, 24.04, amd64 noble image

Architecture

AMI ID

Publish Date

Username

64-bit (x86)

ami-080254318c2d8932f

2026-03-14

ubuntu

Verified provider

▼ Instance type

Info | Get advice

Instance type

t3.micro
Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0108 USD per Hour
On-Demand Windows base pricing: 0.02 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0143 USD per Hour
On-Demand RHEL base pricing: 0.0396 USD per Hour On-Demand SUSE base pricing: 0.0108 USD per Hour

Free tier eligible

All generations

Compare instance types

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 22, 102.89.23.101/32)

Type

Info

ssh

Protocol

Info

TCP

Port range

Info

22

Source type

Info

My IP

Name

Info

Q Add CIDR, prefix list or security group

102.89.23.101/32

Description - optional

Info

e.g. SSH for admin desktop

Remove

▼ Security group rule 2 (TCP, 443, 0.0.0.0/0)

Type

Info

HTTPS

Protocol

Info

TCP

Port range

Info

443

Source type

Info

Anywhere

Source

Info

Q Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional

Info

e.g. SSH for admin desktop

Remove

▼ Security group rule 3 (TCP, 80, 0.0.0.0/0)

Type

Info

HTTP

Protocol

Info

TCP

Port range

Info

80

Source type

Info

Anywhere

Source

Info

Q Add CIDR, prefix list or security group

0.0.0.0/0

Description - optional

Info

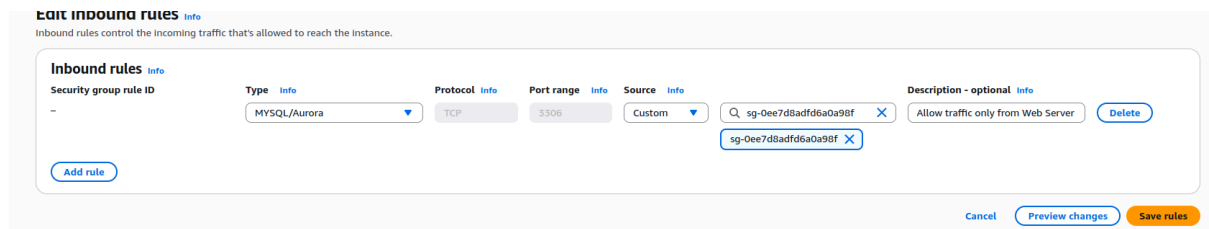
e.g. SSH for admin desktop

Remove

Phase 3: Network Isolation & Security Hardening

In this phase, I implemented strict access control to protect the database tier from external threats.

- **Action:** Modified the RDS Security Group ([Prestashop-DB-SG](#)).
- **Rule Implementation:** Deleted all default public access rules. Created a custom inbound rule allowing **MySQL/Aurora (Port 3306)** traffic strictly from the source Security Group: [Prestashop-Web-SG](#).
- **Result:** The database is mathematically isolated from the public internet, adhering to the Principle of Least Privilege.



Phase 4: System Administration & LAMP Stack Installation

Configured the Ubuntu server via the Linux command-line interface.

1. Secure SSH Connection:

Bash

```
chmod 400 bincom-key.pem
```

```
ssh -i "bincom-key.pem" ubuntu@13.62.103.215
```

2. Environment & Dependency Setup:

Bash

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt install apache2 mysql-client unzip -y
```

```
sudo apt install php libapache2-mod-php php-mysql php-curl php-gd php-mbstring  
php-xml php-zip php-intl php-bcmath -y
```

3. Application Deployment & Permissions:

Bash

```
cd /var/www/html
```

```
sudo rm index.html
```

```
sudo wget https://github.com/PrestaShop/PrestaShop/releases/download/8.2.0/prestashop_8.2.0.zip
```

```
sudo unzip prestashop_8.2.0.zip
```

```
sudo chown -R www-data:www-data /var/www/html/
```

```
sudo chmod -R 755 /var/www/html/
```

```
exploitx@exploitx-ThinkPad-T470-W10D6:~$ cd ~/Downloads
exploitx@exploitx-ThinkPad-T470-W10D6:~/Downloads$ ssh -i "bincom-key.pem" ubuntu@13.62
.103.215
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1007-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Fri Mar 27 13:14:05 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   26.3% of 6.71GB Processes:      110
Memory usage: 23%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.36.30

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-36-30:~$
```

```
prestashop_8.2.0 100%[=====] 78.52M 38.7MB/s in 2.0s
2026-03-27 13:26:22 (38.7 MB/s) - 'prestashop_8.2.0.zip' saved [82333023/82333023]

ubuntu@ip-172-31-36-30:/var/www/html$ sudo unzip prestashop_8.2.0.zip
Archive:  prestashop_8.2.0.zip
  inflating: prestashop.zip
  inflating: index.php
  inflating: Install PrestaShop.html
ubuntu@ip-172-31-36-30:/var/www/html$ sudo chown -R www-data:www-data /var/www/html/
ubuntu@ip-172-31-36-30:/var/www/html$ sudo chmod -R 755 /var/www/html/
ubuntu@ip-172-31-36-30:/var/www/html$ sudo systemctl restart apache2
ubuntu@ip-172-31-36-30:/var/www/html$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
  systemctl restart apache2
ubuntu@ip-172-31-36-30:/var/www/html$ sudo systemctl restart apache2
ubuntu@ip-172-31-36-30:/var/www/html$
```

Phase 5: Application Configuration & Database Linkage

Finalized the connection between the application and data tiers.

1. Apache Optimization:

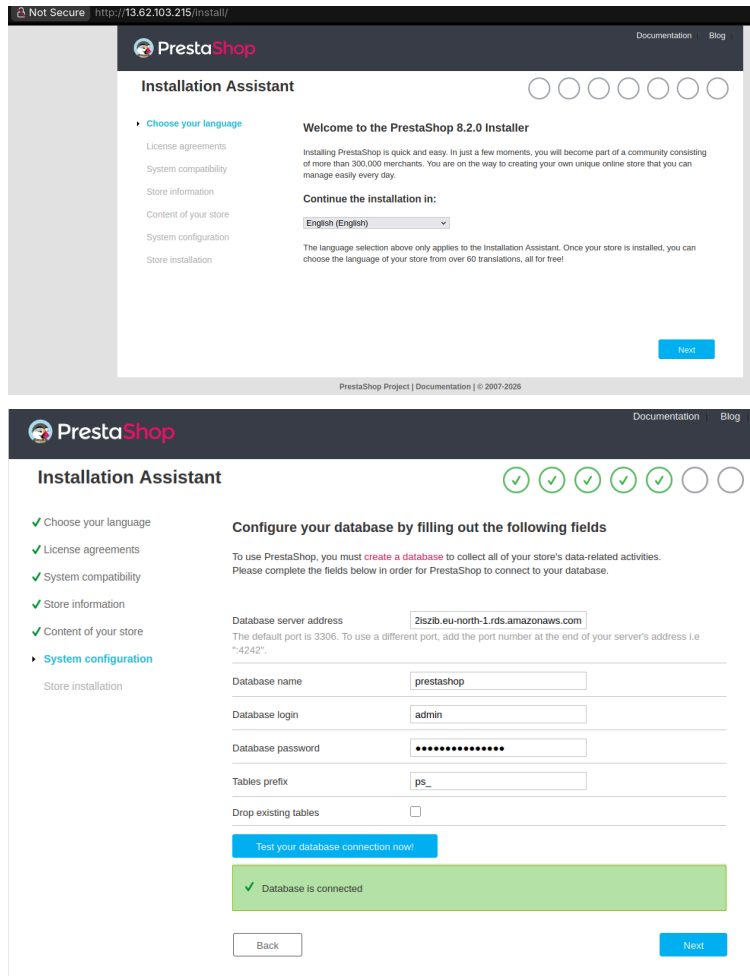
Bash

```
sudo a2enmod rewrite
```

```
sudo systemctl restart apache2
```

2. Database Integration:

- **Database Server:** prestashop-db.cnaqkk2iszib.eu-north-1.rds.amazonaws.com
- **Database Name:** prestashop
- **Database User:** admin
- **Database Password:** [REDACTED FOR SECURITY]



Phase 6: Final Security Audit & Lockdown

Following successful deployment, the server was hardened for production.

1. Installer Cleanup:

Bash

```
sudo rm -rf /var/www/html/install
```

2. Backend Obfuscation: Verified the automated obfuscation of the administrator backend URL to thwart automated bot attacks and brute-force discovery attempts.

- **Admin Login Email:** felixdemkiir@gmail.com
- **Admin Password:** [REDACTED FOR SECURITY]

```
ubuntu@ip-172-31-36-30:~$ sudo rm -rf /var/www/html/install
ubuntu@ip-172-31-36-30:~$
```

Conclusion

The e-commerce platform is fully operational, decoupled, and secured behind strict network firewalls. The architecture ensures that public web traffic can access the store, while the sensitive database remains entirely hidden and accessible only to the internal web server.